



## M002 Week 5 — Mini Cube Puzzle

**Topic:** Mini Cube Puzzle · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

Finish the mini cube. Keep the starter going: add **2 more conditionals** with **one** **OR**, fill the cube, then break a hole in the **top** and climb out.

### 1 · Predict 🧠

Write what happens.

```
for i in range(5):
    agent.place(BLOCK)
    agent.move(FORWARD, 1)
```

What shape is left behind? How long is it?

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```
if on_edge or on_top_layer:
    agent.place(BLOCK)
else:
    pass # leave empty
```

Only one condition must be true. Which spots get a block? Which stay empty?

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## 2 · Trace 🦶

```
for layer in range(3):
    fill_layer()
    agent.move(UP, 1)
if agent.detect(BLOCK, UP):
    # break above, then climb
    agent.move(UP, 1)
```

**How many layers fill? Where does the agent end up? Is there still a block above it?**

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### 3 · Spot + Fix the Bugs 🐛

**Bug A** — The agent should place blocks across the layer, **then** move up. It moves up too early.

```
agent.place(BLOCK)
agent.move(UP, 1)
agent.place(BLOCK)
agent.place(BLOCK)
```

**Fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — Place a block when on the **bottom layer** **OR** on an **outer wall**. It only checks the bottom.

```
if on_bottom_layer:
    agent.place(BLOCK)
```

**Hint:** add the wall check with **or**.

**Fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — To climb out, break the block above **first**, then move up. It moves up into solid block and gets stuck.

```
agent.move(UP, 1)
agent.destroy(UP)
```

**Hint:** make the hole, then move into it.

**Fixed code:**

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**Why was it wrong? Why does your fix work?**

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#### 4 · Write Code 📌

Write a loop that fills **one layer**: place a block, move forward, repeat 4 times.

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Write a conditional that uses **or** : if the agent is on a **corner** **OR** on the **top**, place a block.

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Open this week's world. Find the **mini cube starter** — part is built.

1. Keep the starter running.
2. Add **2 more conditionals** that decide where blocks go.
3. Make **one** use **or** (e.g. corner **OR** top → place glass).
4. Break a hole in the **top** and climb out.

Record **one video** (a phone is fine). Show two things:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

**Write your lines here, then say them in your video.**

**Submit** 

Send this worksheet + your video to teacher on KakaoTalk.